An exact two-LMI reduction of the canonical Q_{222} slack and an obstruction to every polynomial Bob selector

Abstract

A source-independent polynomial state change reduces the complete canonical slack, with all nine source coefficients and four independent state coordinates, to a matrix quadratic in the squared Alice opening. Positivity in Alice is equivalent to two 4×4 matrix inequalities with one 2×2 Bob-dependent selector. This reduction includes all zero source weights and all projective testing endpoints. On C1, every polynomial selector reduces to three scalar parameters. An already certified rational valid source admits none: its exact contact fixes those parameters, and a 3×3 principal minor is negative at $v = 1$. Thus polynomial completion of this normal form fails; a fixed algebraic selector or an additional fixed testing multiplier remains unresolved.

1 The full state reduction

Let $c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \geq 0$ and put $D_u = 1 + u^2$, $D_v = 1 + v^2$. The established complete canonical parent is

$$\begin{aligned} F_c = {} & 4\lambda D_u (W + vX)^2 + 4nD_u D_v X^2 + 4m(vW - X + uvY - uZ)^2 \\ & + 4CD_v (uW - Y)^2 + 4AD_u D_v Y^2 + (4D + 2d)(uW + uvX - Y - vZ)^2 \\ & + (4\alpha + 2d) D_u (vY - Z)^2 + (4\beta + 2d) D_v (X + uZ)^2 - 2dD_u D_v Z^2. \end{aligned} \tag{1}$$

Define the symmetric 2×2 matrices

$$\begin{aligned} P(v) &= \begin{pmatrix} 4mv^2 & -4mv \\ -4mv & 4m + (4\beta + 2d) D_v \end{pmatrix}, \\ N(v) &= \begin{pmatrix} 4CD_v + 4D + 2d & (4D + 2d)v \\ (4D + 2d)v & (4D + 2d)v^2 \end{pmatrix}, \\ E(v) &= \begin{pmatrix} 4AD_v + (4\alpha + 2d)v^2 & -(4\alpha + 2d)v \\ -(4\alpha + 2d)v & 4\alpha - 2dv^2 \end{pmatrix}, \\ K(v) &= 4\lambda \begin{pmatrix} 1 & v \\ v & v^2 \end{pmatrix} + 4nD_v \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \tag{2}$$

All entries are affine in c and quadratic in the testing variable v. For $a = (S, T)^T$, $b = (P_0, Q_0)^T$, and $U = u^2$, make the polynomial change

$$W = S + UP_0, \quad X = T + UQ_0, \quad Y = u(S - P_0), \quad Z = u(T - Q_0). \tag{3}$$

The state coordinates P_0, Q_0 are unrelated to the matrix $P(v)$. Direct substitution gives the exact identity

$$F_c = D_u \Phi_c(U, v; a, b), \qquad \boxed{\begin{aligned} \Phi_c = {} & (1 + U) a^T P a + U(1 + U) b^T N b \\ & + U(a - b)^T E (a - b) + (a + Ub)^T K (a + Ub). \end{aligned}} \tag{4}$$

For example, the m, C, D, α, β event roots in (1) become respectively $D_u(vS - T)$, uD_uP_0, $uD_u(P_0 + vQ_0)$, $u\{v(S - P_0) - (T - Q_0)\}$, and D_uT. These substitutions prove (4) without a source-dependent inverse.

For completeness, write $\psi = (W, X, Y, Z)^T$ and let $\mathcal{P}_c(U, v)$ be the symmetric matrix of Φ_c in (a, b). Its three coefficients are

$$\mathcal{P}_c = \begin{pmatrix} P + K & 0 \\ 0 & 0 \end{pmatrix} + U \begin{pmatrix} P + E & K - E \\ K - E & N + E \end{pmatrix} + U^2 \begin{pmatrix} 0 & 0 \\ 0 & N + K \end{pmatrix}. \tag{5}$$

The change matrix has determinant $u^2 D_u^2$. Its polynomial inverse numerator is

$$L(u) = \begin{pmatrix} u & 0 & u^2 & 0 \\ 0 & u & 0 & u^2 \\ u & 0 & -1 & 0 \\ 0 & u & 0 & -1 \end{pmatrix}, \qquad \boxed{u^2 D_u F_c(\psi) = (L(u)\psi)^T \mathcal{P}_c(u^2, v)(L(u)\psi).} \tag{6}$$

Equation (6) is a polynomial normal-form identity. It is not a sum-of-squares certificate.

2 Two matrix inequalities, with the quantifiers retained

Theorem 1. *For any fixed real source c and fixed real Bob test v, the following are equivalent:*

1. *$\mathcal{P}_c(U, v) \succeq 0$ for every $U \geq 0$.*
2. *There exists a real 2×2 matrix J such that*

$$\boxed{\mathcal{G}_J = \begin{pmatrix} P + K & J \\ J^T & N + K \end{pmatrix} \succeq 0, \qquad \mathcal{H}_J = \begin{pmatrix} P + E & K - E - J \\ K - E - J^T & N + E \end{pmatrix} \succeq 0.} \tag{7}$$

The equivalence also holds over every real closed field. In particular, complete canonical validity is exactly

$$\boxed{\forall v \in \mathbb{R}\ \exists J \in \mathbb{R}^{2\times 2} : \mathcal{G}_J(v) \succeq 0, \quad \mathcal{H}_J(v) \succeq 0.} \tag{8}$$

All source faces, including $d = 0$ and arbitrary zero event weights, are retained.

Proof. The reverse implication is the explicit identity

$$\Phi_c = \begin{pmatrix} a \\ Ub \end{pmatrix}^T \mathcal{G}_J \begin{pmatrix} a \\ Ub \end{pmatrix} + U \begin{pmatrix} a \\ b \end{pmatrix}^T \mathcal{H}_J \begin{pmatrix} a \\ b \end{pmatrix}. \tag{9}$$

For the forward implication, $\mathcal{P}_c(t^2, v)$ is a positive semidefinite univariate matrix polynomial of degree at most four. The real-line matrix Fejér–Riesz theorem gives a polynomial matrix factor of degree at most two; real and imaginary parts give a real sum of matrix squares [1, Theorem 1.2]. Averaging its values at t and $-t$ separates even and odd powers. Consequently

$$\mathcal{P}_c(U, v) = (B_0 + UB_2)^T (B_0 + UB_2) + UB_1^T B_1$$

for rectangular real matrices B_0, B_1, B_2. The constant coefficient in (5) has its last two rows and columns zero, so $B_0 = [R \ \ 0]$. The coefficient of U^2 has its first two rows and columns zero, so $B_2 = [0 \ \ T]$. This uses only the elementary fact that a zero diagonal entry of a positive semidefinite Gram matrix forces its entire row and column to vanish. Setting $J = R^T T$ gives $\mathcal{G}_J = [R\ T]^T [R\ T]$ and $\mathcal{H}_J = B_1^T B_1$, proving (7).

This is a fixed finite-dimensional first-order assertion in the real closed field language: positivity is expressed by principal minors, and the half-line condition by quantification over U and a four-vector. Transfer therefore proves precisely the same assertion over every real closed field.

For $u \neq 0$, (3) is invertible and so positivity of F_c is equivalent to positivity of $\mathcal{P}_c(u^2, v)$. The excluded value $u = 0$ is recovered by polynomial continuity; all $U > 0$ are attained. More explicitly, (1) has bidegree at most $(2, 2)$ in (u, v), so

$$F_c^{\mathrm{h}}(r, s; \psi) = r_0^2 s_0^2 F_c(r_1/r_0, s_1/s_0; \psi)$$

is polynomial in the homogeneous angle pairs $r = (r_0, r_1)$ and $s = (s_0, s_1)$. On their unit circles it equals the normalization $F_c/(D_u D_v)$. Continuity recovers the Alice and Bob projective endpoints. Over a real closed field the same endpoint statement follows by the sign of the first nonzero coefficient in the corresponding local polynomial chart. No source coefficient was divided out. □

There is also an elementary check on the face $d = 0$: then $E \succeq 0$, and $J = K$ makes both matrices in (7) sums of the positive diagonal blocks P, N and a square block built from K or E. More generally adding a nonnegative event source with $d = 0$ preserves any certificate by adding its K to J. The unresolved completion is therefore associated with the signed core, not with adding positive events.

Theorem 1 does not choose J in a fixed space of functions of v. Its real-closed-field proof permits coefficients algebraic in both source and testing data. The tensor product of the independent source and testing fields is not a real closed field, and this proof supplies no sum-of-squares factorization in that tensor product.

3 The complete polynomial-selector obstruction

Proposition 2. *If polynomial entries $J(v)$ satisfy (7) for every real v, they have degree at most two. A selector can then be chosen of the form*

$$J(v) = \begin{pmatrix} p_0 + p_2 v^2 & q_1 v \\ r_1 v & s_0 + s_2 v^2 \end{pmatrix}.$$

On the entire C1 face $\lambda = n = 0$, existence of any polynomial selector is equivalent to existence of one with just three scalar parameters:

$$\boxed{J(v) = \begin{pmatrix} pv^2 & 0 \\ rv & sv^2 \end{pmatrix}.} \tag{10}$$

Proof. Every 2×2 principal minor of $\mathcal{G}_J$ gives $|J_{ij}(v)|^2 \leq (P + K)_{ii}(N + K)_{jj}$. The right side has degree at most four, proving the degree bound, including cases where it is identically zero. With $T = \mathrm{diag}(1, -1)$, each matrix in (2) obeys $M(-v) = TM(v)T$. Replace $J(v)$ by $\frac{1}{2}\{J(v) + TJ(-v)T\}$; convexity of the positive semidefinite cone retains both inequalities and gives the displayed parity.

On C1, $P(0) = \mathrm{diag}(0, 4m + 4\beta + 2d)$ and $N(0) = \mathrm{diag}(4C + 4D + 2d, 0)$. Zero diagonal entries in $\mathcal{G}_J(0)$ force $p_0 = s_0 = 0$. Finally

$$q_1^2 v^2 = |J_{12}(v)|^2 \leq 4m(4D + 2d)v^4$$

for every v, so $q_1 = 0$. This proof does not assume any of the source weights is positive. □

Use now the already established valid C1 source from `strict_outer_physical_source.json` in the public repository, with $d = 1$, $\lambda = n = 0$, and

$$m = \frac{1166523000}{31661820415017653}, \qquad C = \frac{322803000}{186199784281}, \qquad A = \frac{218288946200}{1359122513},$$
$$D = \frac{90943434293}{222507771414}, \qquad \alpha = \frac{248802430236}{3300726103}, \qquad \beta = \frac{205476973}{51646655494}.$$

Its validity is an established input from [2], certified by the independently checked order-13744 rational certificate; that certificate is not rebuilt here. Its exact contact is

$$u_* = \frac{1}{13}, \qquad v_* = 16, \qquad (W_*, X_*, Y_*, Z_*) = (140, -1/32, 1/50, 1).$$

In (3) the contact becomes

$$U_* = \frac{1}{169}, \quad a_* = \begin{pmatrix} 69589/500 \\ 247/5440 \end{pmatrix}, \qquad b_* = \begin{pmatrix} 69459/500 \\ -70473/5440 \end{pmatrix}. \tag{11}$$

The full four-dimensional equation $\mathcal{P}_c(U_*, v_*)(a_*, b_*)^T = 0$ is exact.

Theorem 3. *This valid source has no polynomial selector satisfying* (7) *for all* v*. Equivalently, its polynomial* $\Phi_c(u^2, v; a, b)$ *is not a sum of polynomial squares in* (u, v, a, b)*.*

Proof. By Proposition 2, a polynomial selector would have the form (10). At the contact, both nonnegative terms in (9) vanish. Therefore

$$J(v_*)b_* = -P(v_*)a_*/U_*, \qquad J(v_*)^T a_* = -U_* N(v_*)b_*. \tag{12}$$

The first coordinate of the first equation determines p, and the second coordinate of the second equation determines s; the remaining coordinates then determine r consistently. Substituting the displayed rational input gives

$$p = -\frac{37168625}{1489598274248}, \qquad r = \frac{622145600000}{505399414477}, \qquad s = \frac{7465747200}{3689046821}. \tag{13}$$

This is a rank-three linear system, not a numerical fit.

Put $a_0 = 4m$, $b_0 = 4\beta + 2d$, $c_0 = 4C$, $e_0 = 4D + 2d$. At $v = 1$, the principal submatrix of $\mathcal{G}_J$ on coordinates (a_2, b_1, b_2) is

$$\begin{pmatrix} a_0 + 2b_0 & r & s \\ r & 2c_0 + e_0 & e_0 \\ s & e_0 & e_0 \end{pmatrix}.$$

Its determinant equals

$$2e_0c_0(a_0 + 2b_0) - e_0(r - s)^2 - 2c_0s^2. \tag{14}$$

The input fractions imply directly

$$3 < e_0 < 4, \quad 0 < c_0 < 1/100, \quad 0 < a_0 + 2b_0 < 5, \quad r < 5/4, \quad s > 2.$$

Consequently (14) is less than $2/5 - 27/16 = -103/80 < 0$, contradicting $\mathcal{G}_J(1) \succeq 0$.

Finally, a polynomial sum of squares for $\Phi_c(u^2, v; a, b)$ can use roots of degree at most two in u, at most one in v, and homogeneous linear degree in the four state coordinates: higher terms cannot cancel in a sum of real squares. Averaging in $u \mapsto -u$ and using the constant and leading state supports exactly as in Theorem 1 would produce a polynomial selector. Conversely a polynomial selector gives polynomial matrix factorizations of its two positive univariate matrices by Fejér–Riesz and then a polynomial sum of squares through (9). This proves the equivalence. □

4 What remains, and the evaluated files

The reduction is exact for the whole canonical slack cone, with both state normals and every zero opening retained. Its evaluated block diagonal order-eight pencil $\mathcal{G}_J \oplus \mathcal{H}_J$ has 108 integer COO entries, nine source parameters, four selector parameters, and Bob degree at most two. The derived C1 three-parameter version has 80 integer COO entries. They are *pointwise pencils*: the quantifier $\forall v\, \exists J$ remains.

The obstruction is to every polynomial Bob selector on the entire real Bob chart in this normal form, not to every source-independent algebraic family or multiplier. It also does not show that $u^2 D_u F_c$ in the original state coordinates has no polynomial sum-of-squares representation: cancelling the extra factors after pullback through (3) would require another theorem. A viable completion must supply one fixed testing algebra (for example one fixed Bob denominator or a genuinely algebraic family) in which the selectors and their positive matrix factors exist uniformly over the full source cone. The pointwise real-closed-field theorem and source-dependent matrix factorization do not supply that assertion. In particular, an odd algebraic function such as $v|v|$ can vanish quadratically at $v = 0$ without vanishing identically, so the rank argument forcing the polynomial entry $J_{12} = 0$ does not extend to a two-half-line algebraic cover. No adequacy statement for that cover has been proved here.

A concrete sufficient remaining implication is the existence of one fixed nonnegative polynomial $q(v)$, positive on a dense set, for which every valid source admits a polynomial $Z_c(v)$ satisfying

$$\begin{pmatrix} q(P+K) & Z_c \\ Z_c^T & q(N+K) \end{pmatrix} \succeq 0, \qquad \begin{pmatrix} q(P+E) & q(K-E)-Z_c \\ q(K-E)-Z_c^T & q(N+E) \end{pmatrix} \succeq 0 \quad (v \in \mathbb{R}).$$

The first inequality would force $\deg Z_c \le \deg q + 2$; univariate matrix factorization would then give one fixed testing family for $q(v)u^2 D_u F_c$, with source-linear coefficient input and an exact validity converse by density. Theorem 3 rules out $q = 1$ for this route. It neither proves nor disproves the displayed implication for another q, or the broader possibility of a fixed algebraic family. No full-set affine Q_{222} lift is claimed here.

The two integer COO files accompanying this note contain respectively the pointwise order-eight pencil and its derived C1 polynomial-selector version. The counterexample JSON file contains the exact contact linear system, its forced solution, and the negative minor. An independent verifier checks every exported matrix coefficient and the counterexample using only Python's standard-library rational arithmetic. A separate symbolic checker verifies the full nine-source polynomial identities. The supplied builder recreates only this new data; the README gives the exact commands.

References

[1] A. Zalar, *Matrix Fejér–Riesz theorem with gaps*, Theorem 1.2, arXiv:1503.06034; author's full text.

[2] A. Chaturvedi, *q222-finite-sdp*, commit 5fc0a0487bb9. Established input: `extracted-lifts/outer-candidates/ strict_outer_physical_source.json`; the exact file is bundled with the new certificate data.